ESP32-Based Healthcare Monitoring System with ECG Analysis, Edge AI, and Fall Detection

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***Abstract*—This paper presents a low-cost, portable healthcare monitoring system built around an ESP32 microcontroller for round-the-clock tracking of key physiological parameters and early detection of medical emergencies. The system integrates an AD8232 ECG sensor for cardiac rhythm analysis, a MAX30100 sensor for blood oxygen saturation (SpO2), an NTC thermistor for body temperature, and an MPU6050 accelerometer for fall detection. ECG signals are processed to extract RR intervals, which are classified by a logistic regression model trained in Google Colab and deployed on the ESP32 as Normal, Tachycardia or Bradycardia. All readings and predictions are sent over the ESP32's built-in Bluetooth to a smartphone running a Serial Bluetooth Terminal app. The device is powered by a 2-cell Li-ion battery and a buck converter providing 5 V. By performing inference locally, the system avoids cloud dependence and preserves privacy, making it suitable for home use, elderly care and telemedicine.**

***Index Terms*—ESP32, ECG, edge AI, logistic regression, tachycardia, bradycardia, SpO2, fall detection, MPU6050, healthcare monitoring, Bluetooth.**

I. INTRODUCTION

Cardiovascular diseases (CVDs) remain the leading cause of death globally, claiming an estimated 17.9 million lives each year according to the World Health Organization [1]. Early detection of abnormal heart rhythms such as tachycardia (heart rate above 100 BPM) and bradycardia (below 60 BPM) is important for timely intervention. Falls are a second major risk: roughly 28–35% of people aged 65 and above experience at least one fall annually, often resulting in fractures, hospitalization and reduced quality of life [2].

Conventional monitoring depends on periodic clinic visits and expensive equipment, which limits continuous observation and delays detection of transient events. The availability of low-cost microcontrollers, biomedical sensors and machine-learning tools now makes affordable wearable monitors feasible. This work develops a single portable device that combines cardiac rhythm analysis, SpO2 monitoring, body temperature measurement and fall detection. The ESP32 is well suited to this role: it provides a dual-core processor, built-in Bluetooth and Wi-Fi, multiple ADC channels and low power consumption [3], so edge inference can run without cloud connectivity, giving low latency and keeping data local.

Contributions: (i) an integrated four-sensor monitoring node on one ESP32; (ii) RR-interval-based cardiac classification using a logistic regression model whose parameters are embedded in firmware; (iii) Bluetooth reporting to a smartphone without internet; (iv) a battery-powered design intended for home and elderly-care use; and (v) a documented evaluation plan with metrics for future clinical-grade validation.

The remainder of this paper is organized as follows. Section II reviews related work. Section III states the problem and objectives. Sections IV–VI describe the architecture, hardware and methodology. Sections VII–IX cover power analysis, evaluation plan and discussion, and Section X concludes.

II. RELATED WORK

A. ECG Analysis and QRS Detection

Real-time QRS detection is a well-established topic. The Pan–Tompkins algorithm applies band-pass filtering, differentiation, squaring and moving-window integration with adaptive thresholds to locate R-peaks reliably, and has become the reference for embedded implementations [4]. The RR interval derived from successive R-peaks is also the basis of heart-rate-variability analysis, for which standards of measurement were defined by a joint task force. This work uses a lightweight filter-and-threshold variant suited to a microcontroller.

B. Machine Learning on Microcontrollers

Running inference directly on small devices (often called TinyML or edge AI) reduces latency, bandwidth and privacy exposure compared with cloud processing [5], [6]. Simple models such as logistic regression are attractive here because their parameters (a weight and bias per class) can be exported from a training environment and evaluated with a few multiplications and an exponential, which is trivial for a 240 MHz processor.

C. Wearable Fall Detection

Threshold-based fall detection with tri-axial accelerometers is widely used because it is computationally cheap. Such methods typically look for a drop in acceleration magnitude during free-fall followed by a large impact peak, optionally confirmed by posture or inactivity [7]. The main weakness is false positives from vigorous daily activities, which motivates tunable thresholds and confirmation stages.

D. Positioning of This Work

Many existing monitors address one parameter, rely on cloud services, or are costly [8], [9]. The present design combines four measurements on a single low-cost ESP32 node, performs classification locally, and reports over Bluetooth. The novelty of this work is system-level integration and the embedded deployment of a compact classifier, not a claim of improved clinical accuracy.

III. PROBLEM STATEMENT AND OBJECTIVES

A. Problem Statement

- CVDs require early detection of arrhythmias; tachycardia and bradycardia can be identified from heart-rate patterns.
- Traditional ECG monitoring needs clinical visits and costly equipment, limiting ongoing health tracking.
- Falls among the elderly cause serious injuries and often go unreported.

- SpO2 and body temperature are vital signs that give important health insight.
- Existing monitors are expensive, bulky or cloud-dependent; edge AI can give real-time, privacy-preserving assessment without internet.

B. Objectives

- Design an ESP32-based monitor with multiple biomedical sensors.
- Acquire ECG and extract RR intervals using digital signal processing.
- Train a logistic regression model in Google Colab (Normal, Tachycardia, Bradycardia) and deploy it on the ESP32 for real-time inference.
- Measure SpO2 (MAX30100), body temperature (NTC thermistor, Steinhart–Hart) and detect falls (MPU6050).
- Transmit all data to a mobile app over Bluetooth and power the system from a 2-cell Li-ion battery with a 5 V buck converter.

IV. SYSTEM ARCHITECTURE

The ESP32 coordinates data acquisition, edge inference and wireless transmission. The ECG sensor and NTC thermistor connect to analog input pins, while the MAX30100 and MPU6050 connect over the shared I²C bus. A 2-cell Li-ion battery (7.4 V nominal) with a buck converter supplies a stable 5 V; the 3.3 V rail required by the AD8232 and MAX30100 is obtained from the ESP32 development board's on-board regulator. The firmware continuously samples all sensors and sends formatted data to the mobile app at 1 Hz. Fig. 1 shows the overall block diagram.

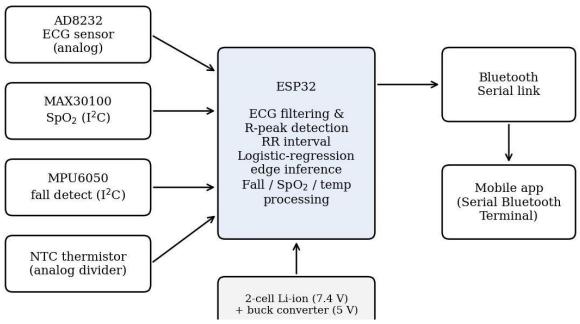


Fig. 1: Block diagram of the proposed healthcare monitoring system.

The firmware follows the sequence in Fig. 2. ECG sampling is time-critical, so it is driven by a 250 Hz timer, while slower quantities (SpO2, temperature, motion summary) are handled in the main loop. Because the ESP32 has two cores, acquisition and Bluetooth communication can be placed on separate tasks so that radio activity does not disturb ECG timing.

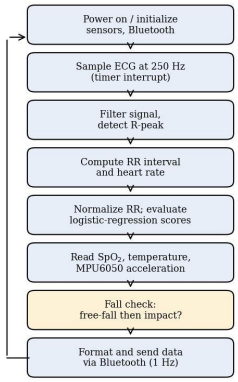


Fig. 2: Firmware flowchart.

V. HARDWARE DESCRIPTION

A. ESP32 Controller

The ESP-WROOM-32 module is built around the ESP32 chip, a 2.4 GHz Wi-Fi and Bluetooth combo device. It has two CPU cores with clock frequency adjustable from 80 MHz to 240 MHz, 520 KB SRAM and 4 MB flash, and supports UART, I²C, SPI and ADC peripherals [3]. Its on-board Bluetooth is used to reach the smartphone.

B. AD8232 ECG Module

The AD8232 is an integrated signal-conditioning block that extracts, amplifies and filters small biopotential signals in the presence of noise from motion or remote electrode placement [10]. The module operates at 3.3 V, provides an analog output, leads-off detection (LO+ and LO−) and a shutdown pin, and connects to electrodes through a 3.5 mm jack using the RA, LA and RL leads. The leads-off outputs are monitored in firmware so that a detached electrode is reported instead of being misread as a flat-line rhythm. It is not a medical-grade device.

C. MAX30100 Pulse Oximeter

The MAX30100 measures absorbance of pulsing blood using a red and an infrared LED with a photodetector, and communicates over I²C at a 3.3 V supply. Readings can be degraded by motion, temperature changes and excessive finger pressure. In this design only the SpO2 value is used; heart rate is taken from the ECG to avoid redundancy.

D. NTC Thermistor

A 10 kΩ NTC thermistor in a voltage divider with a 10 kΩ fixed resistor is read by an ADC pin. The resistance is converted to temperature in Celsius using the Steinhart–Hart equation.

E. MPU6050 Motion Sensor

The MPU6050 combines a 3-axis gyroscope and a 3-axis accelerometer with 16-bit ADC output and an on-chip Digital Motion Processor, and communicates over I²C. It is used for fall detection.

F. Power Supply

A 2-cell Li-ion pack (7.4 V, 2000–5000 mAh) feeds an LM2596-based buck converter. The LM2596 accepts 3–40 V input, offers an adjustable 1.5–35 V output, delivers a rated 2 A (3 A maximum with heatsink) and switches at 150 kHz. The output is set to 5 V for the ESP32 and sensors; the module's potentiometer must be adjusted and verified with a multimeter

before connecting the ESP32. The assembled prototype is shown in Fig. 3.

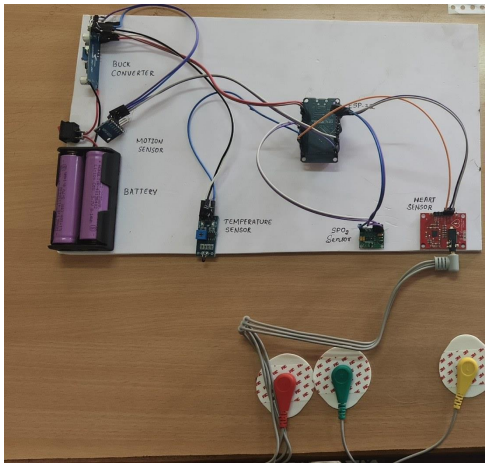


Fig. 3: Hardware prototype of the proposed ESP32-based healthcare monitoring system.

TABLE I: Hardware components

Component	Role
ESP32 (WROOM-32)	Processing, edge AI, Bluetooth
AD8232 + electrodes	ECG acquisition
MAX30100	SpO2 measurement
NTC 10 kΩ + 10 kΩ	Body temperature
MPU6050	Fall detection
2-cell Li-ion, LM2596	Portable 5 V supply
Wearable enclosure	Chest strap or armband

VI. METHODOLOGY

A. ECG Acquisition and RR Extraction

The ECG is sampled at 250 Hz, giving a time resolution of 4 ms per sample, consistent with the ±5 ms RR accuracy target. The raw signal is band-pass filtered to suppress baseline wander and high-frequency noise, then emphasized (differentiated and squared) so that the steep QRS complex dominates. An adaptive threshold marks a sample as an R-peak when it exceeds the threshold and a refractory period (for example 200 ms) has elapsed since the previous peak, which prevents double detection of a single beat. The RR interval and heart rate are then

$$RR_n = t_n - t_{(n-1)}, \quad HR = 60000 / RR_n \tag{1}$$

with RR in milliseconds and HR in BPM. A 600 ms interval therefore corresponds to 100 BPM and a 1000 ms interval to 60 BPM, which are the clinical boundaries for tachycardia and bradycardia used to label the training data.

B. Dataset and Labeling

The training set consists of RR intervals labeled Normal, Tachycardia or Bradycardia (synthetic or clinical). Labels follow the rate definitions above: Tachycardia for RR below about 600 ms, Bradycardia for RR above about 1000 ms, and Normal in between. The data are split into training and test subsets so that performance is measured on unseen intervals.

C. Logistic Regression Model

The model is trained in Google Colab with scikit-learn using one-vs-rest logistic regression with three classes. The single feature, the RR interval, is normalized with the training mean μ and standard deviation σ:

$$z = (RR - \mu) / \sigma \tag{2}$$

For each class k the model learns a weight w_k and intercept b_k, giving a score and probability

$$s_k = w_k z + b_k, \quad p_k = 1 / (1 + e^{(-s_k)}) \tag{3}$$

The reported class is the one with the largest score, k* = argmax s_k. Using the arg-max of the three linear scores allows the middle (Normal) class to win between the two outer classes even though each one-vs-rest sigmoid is monotonic in z. The extracted parameters (μ, σ, a 3×1 weight vector and 3×1 intercepts) are written as constants in the firmware, so no floating-point library beyond exp() is required and inference takes under 10 ms.

D. SpO2 Estimation

The MAX30100 returns red and infrared photoplethysmography (PPG) signals. Oxygen saturation is estimated from the ratio of ratios

$$R = (AC_{red} / DC_{red}) / (AC_{ir} / DC_{ir}) \tag{4}$$

and mapped to SpO2 by an empirical, device-specific calibration curve, commonly approximated as a linear function SpO2 ≈ A − B·R. The constants must be calibrated against a reference oximeter, which is why the ±2% accuracy target is stated for the calibrated case.

E. Body Temperature

The thermistor voltage V is read by the 12-bit ADC (0–4095 counts for 0–3.3 V). With the thermistor on the ground side of the divider, its resistance is $R_{th} = R_f \cdot V/(V_{ref} - V)$, with R_f = 10 kΩ. Temperature T in kelvin is obtained from the Steinhart–Hart equation:

$$1/T = A + B \cdot \ln(R_{th}) + C \cdot (\ln R_{th})^3 \tag{5}$$

where A, B and C are coefficients for the thermistor (or the simpler β-parameter form may be used). Averaging several ADC samples reduces noise, and the result is converted to °C by subtracting 273.15.

F. Fall Detection

The acceleration vector magnitude is computed from the three accelerometer axes:

$$|a| = \sqrt{(a_x^2 + a_y^2 + a_z^2)} \tag{6}$$

At rest |a| ≈ 1 g. During a fall the body first experiences free-fall, where |a| drops well below 1 g, followed within a short time by an impact where |a| rises sharply (Fig. 4). A fall is flagged when a free-fall condition (|a| below a lower threshold) is followed by an impact condition (|a| above an upper threshold) inside a time window of a few hundred milliseconds. Example starting values are 0.4 g and 2.5 g, but these must be tuned experimentally for the wearing position (chest or arm). An optional confirmation stage checks for inactivity or a change in orientation after impact to reduce false alarms from sudden but harmless movements [7].

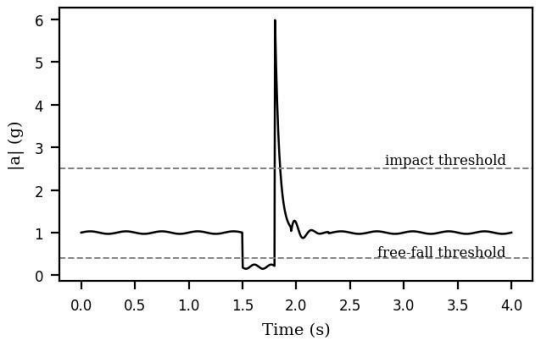


Fig. 4: Illustrative acceleration-magnitude pattern of a fall (schematic, not measured data).

G. Data Format and Transmission

The ECG-derived heart rate, RR interval, cardiac class, SpO2, temperature and fall status are formatted as a text line and sent at 1 Hz through the ESP32's BluetoothSerial interface to the Serial Bluetooth Terminal app on an Android phone, allowing monitoring by the user, family or caregivers. A fall event is sent immediately rather than waiting for the next 1 Hz update, and an out-of-range alert flag can be appended to any abnormal value.

VII. SOFTWARE AND POWER ANALYSIS

A. Software Requirements

Firmware is written in Embedded C/C++ using the Arduino IDE with the ESP32 board package. Libraries used are BluetoothSerial and Wire (both built-in), a MAX30100 library, and an MPU6050 library (e.g., Adafruit_MPU6050). Model training uses Google Colab with Python, scikit-learn, numpy and pandas. The Serial Bluetooth Terminal app displays the received data.

B. Battery Runtime Estimate

Runtime can be estimated from the battery energy, converter efficiency η and average load power:

$$t = (C \cdot V_bat \cdot \eta) / P_load \quad (7)$$

As a worked example, assume an average load of 150 mA at 5 V (P_load = 0.75 W, an assumption that includes the ESP32 with Bluetooth active at roughly 80 mA as listed in its specification plus the sensors) and η = 0.85. A 2000 mAh pack at 7.4 V stores 14.8 Wh, giving $t \approx 14.8 \times 0.85 / 0.75 \approx 16.8$ h, while a 5000 mAh pack stores 37 Wh and gives about 42 h. Actual runtime must be measured, because Bluetooth transmission peaks and LED currents raise the average. Duty-cycling the MAX30100 LEDs and lowering the clock frequency can extend runtime.

VIII. PERFORMANCE METRICS AND EVALUATION PLAN

Table II lists the design specifications of the system. These are targeted values; formal clinical validation has not been performed.

TABLE II: Design performance metrics

Parameter	Value
ECG sampling rate	250 Hz
Heart rate range	30–250 BPM
RR interval accuracy	±5 ms
SpO2 accuracy	±2% (calibrated)
Temperature accuracy	±0.5 °C
Fall detection response	< 1 s
ML inference time	< 10 ms
Data update rate	1 Hz
Bluetooth range	Up to 10 m

To verify these specifications, a structured evaluation is proposed. The classifier is evaluated on a held-out test set using a confusion matrix, from which accuracy, precision, recall and F1-score are computed for each class:

$$Precision = TP/(TP+FP), \quad Recall = TP/(TP+FN) \quad (8)$$

$$F1 = 2 \cdot Precision \cdot Recall / (Precision + Recall) \quad (9)$$

Recall is particularly important for the abnormal classes, since a missed tachycardia or bradycardia is more costly than a false alert.

TABLE III: Proposed verification plan

Subsystem	Reference / method	Metric
RR interval	Compare with a reference ECG recorder	Error in ms
Classifier	Held-out test set	Accuracy, F1
SpO2	Commercial pulse oximeter	Error in %
Temperature	Clinical thermometer	Error in °C
Fall detection	Scripted falls and daily activities	Sensitivity, false alarms
Latency	Timestamps on event and app	Seconds
Battery	Continuous run to cutoff	Hours

IX. DISCUSSION

A. Advantages

- Comprehensive monitoring of cardiac rhythm, SpO2, temperature and falls in one device.
- Cloud-independent edge AI with real-time classification and local data (privacy-preserving).
- Portable, battery-powered and cost-effective; open-source and customizable.
- Early identification of tachycardia and bradycardia; automatic fall detection.

B. Limitations

- ECG quality is sensitive to motion artifacts and electrode placement.
- The model uses only the RR interval, which limits accuracy; it cannot recognize other arrhythmias such as atrial fibrillation.
- Because the label is effectively a heart-rate range, a simple threshold rule would perform similarly; the benefit of the learned model is a probabilistic output and a template for richer features.
- The app shows numeric values only, not the ECG waveform; there is no data storage (SD or cloud).
- No SMS/GSM emergency alert; Bluetooth range is about 10 m.
- Abrupt movements may cause false fall alarms.
- Not a medical-grade device; limited clinical validation, for informational use only.

C. Safety and Ethical Considerations

Because the device handles health data and may be worn by vulnerable users, several precautions apply. The system should clearly state that it is an informational tool and not a diagnostic device, a notice that also appears in the AD8232 manufacturer's documentation [10]. Battery-powered operation avoids mains connection to the body, and the pack should include protection against over-discharge, over-current and short circuit. Since data stay on the local Bluetooth link, privacy risk is lower than for cloud systems, but pairing should be secured so that nearby devices cannot read the stream.

D. Applications

The system can be used for home health monitoring, elderly care (fall detection and physiological readings), telemedicine and remote patient monitoring, post-surgery recovery, sports and fitness, chronic disease management, health research data collection, and portable monitoring in rural areas with limited clinical access.

X. CONCLUSION AND FUTURE WORK

This paper presented an ESP32-based healthcare monitoring system that integrates ECG-based cardiac classification (Normal, Tachycardia, Bradycardia) using logistic regression edge AI, SpO2 monitoring with the MAX30100, temperature measurement with an NTC thermistor and fall detection with the MPU6050. All data are streamed over Bluetooth to a mobile app, and a 2-cell Li-ion battery with a buck converter provides portable 5 V power. The edge-AI approach removes cloud dependence, supporting privacy and low latency.

Future work includes GSM-based emergency SMS alerts, SD-card data logging, Wi-Fi cloud synchronization, additional sensors (blood pressure, glucose), multi-feature and deep-learning models for higher accuracy, ECG waveform display, and integration with electronic health records. Measured results from the evaluation plan in Section VIII will be reported in follow-up work.

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